Homework 4

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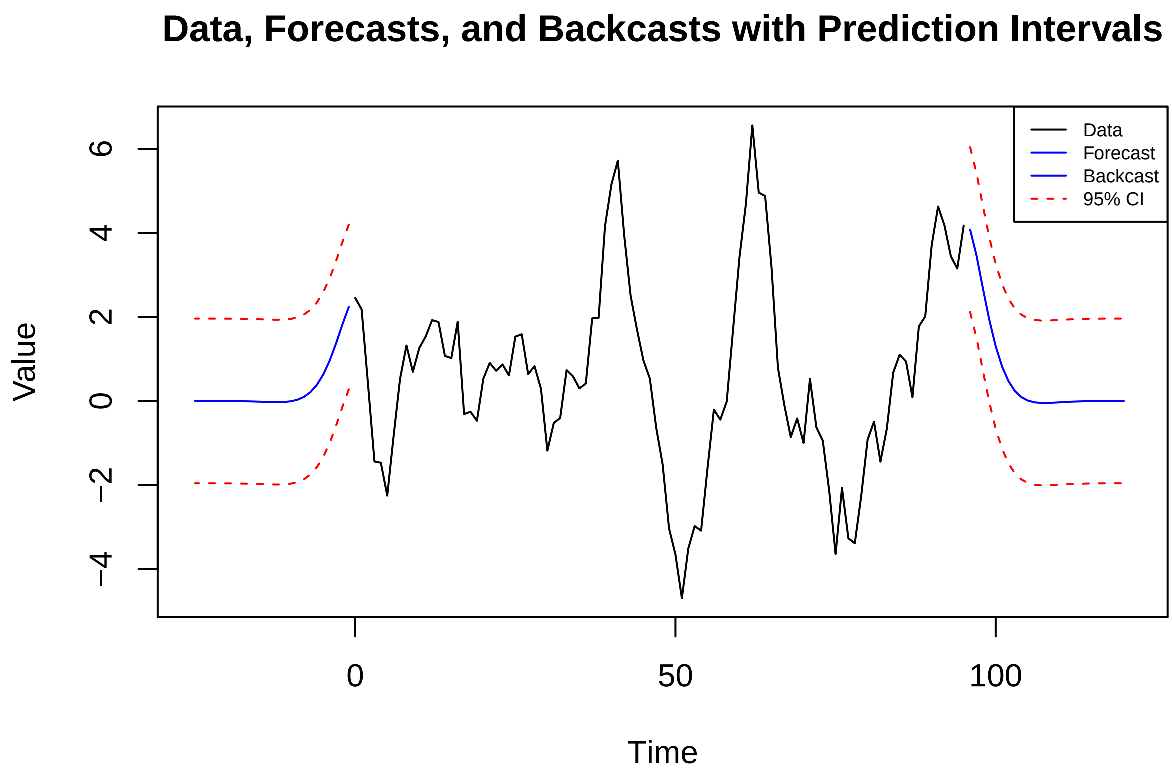
Problem 1



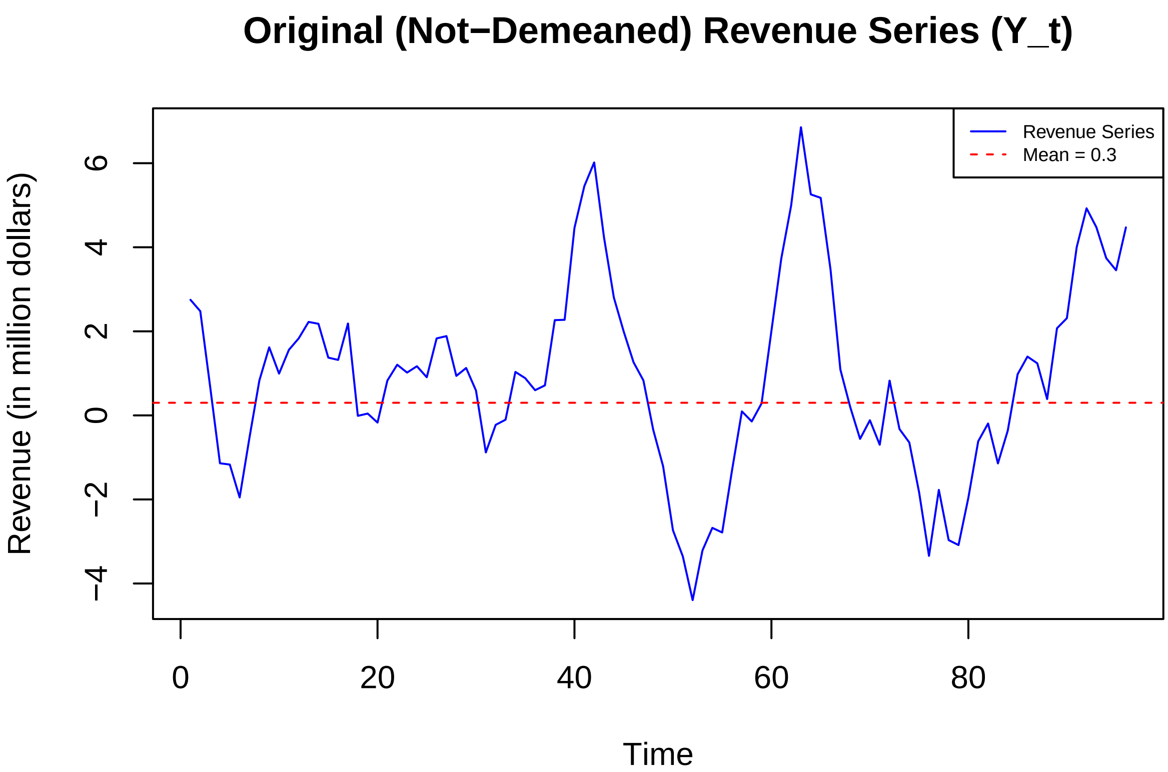
Part b

The 100 steps ahead forecast will be approximately equal to 0 because this is the mean of a demeaned series and the prediction approaches the mean over a very long time horizon like this.

Part c

With no data to base the prediction on, the predicted value would be 0, the mean of a demeaned series, and the Mean Squared prediction error (E) would be 1, the variance of the white noise. E is greater than P_{97}^{96} because P_{97}^{96} is based on data and is not far in the future from the data so it's more accurate.

Part d

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## Prediction for Y at time n+1: 4.378376
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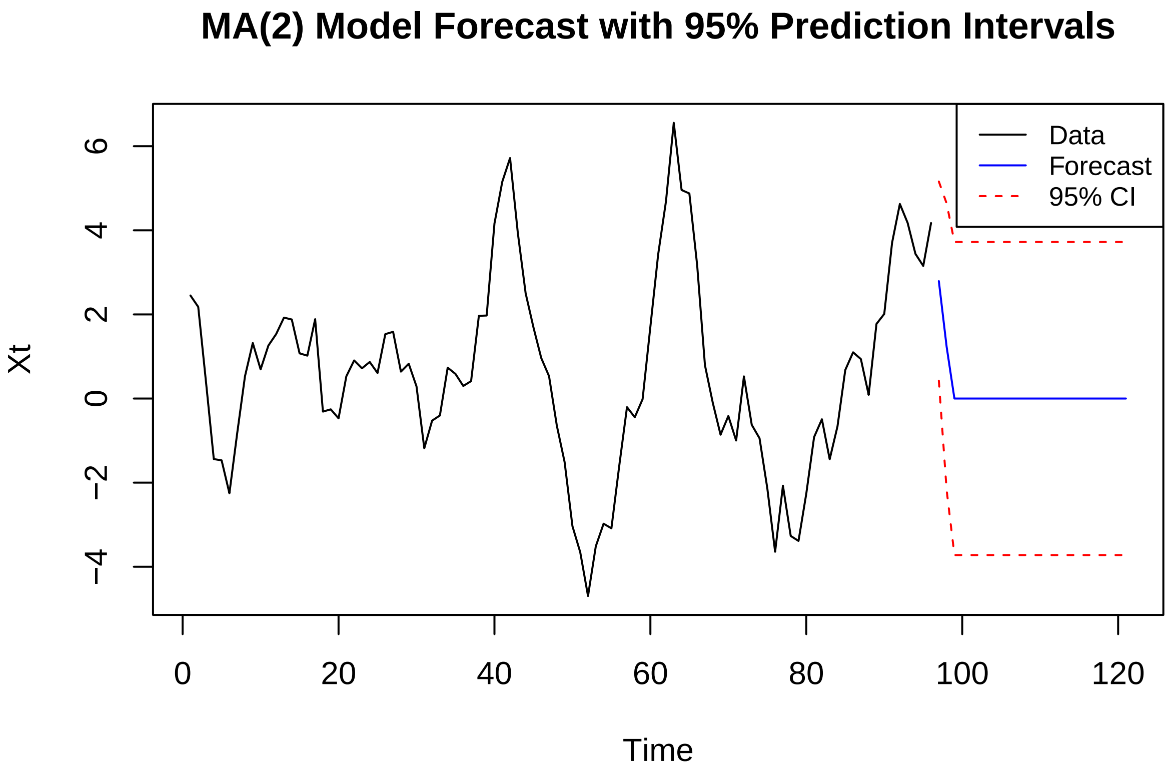
Prediction equation: $Y_{n+1}^{n} = \mu + \phi_1(Y_n - \mu) + \phi_2(Y_{n-1} - \mu)$

Part e

They should make the investment because the estimate of their revenue is \$4.378 million and even if the standard error of this estimate was the variance of the white noise (in this case 1), which in reality it must be less than because it's based on data as mentioned in part c, the lower bound of the 95% confidence interval would be 4.378 - 1.96 = \$2.418 which is much larger than \$1.1 million so the company executives can be confident that they will not go bankrupt.

Part f



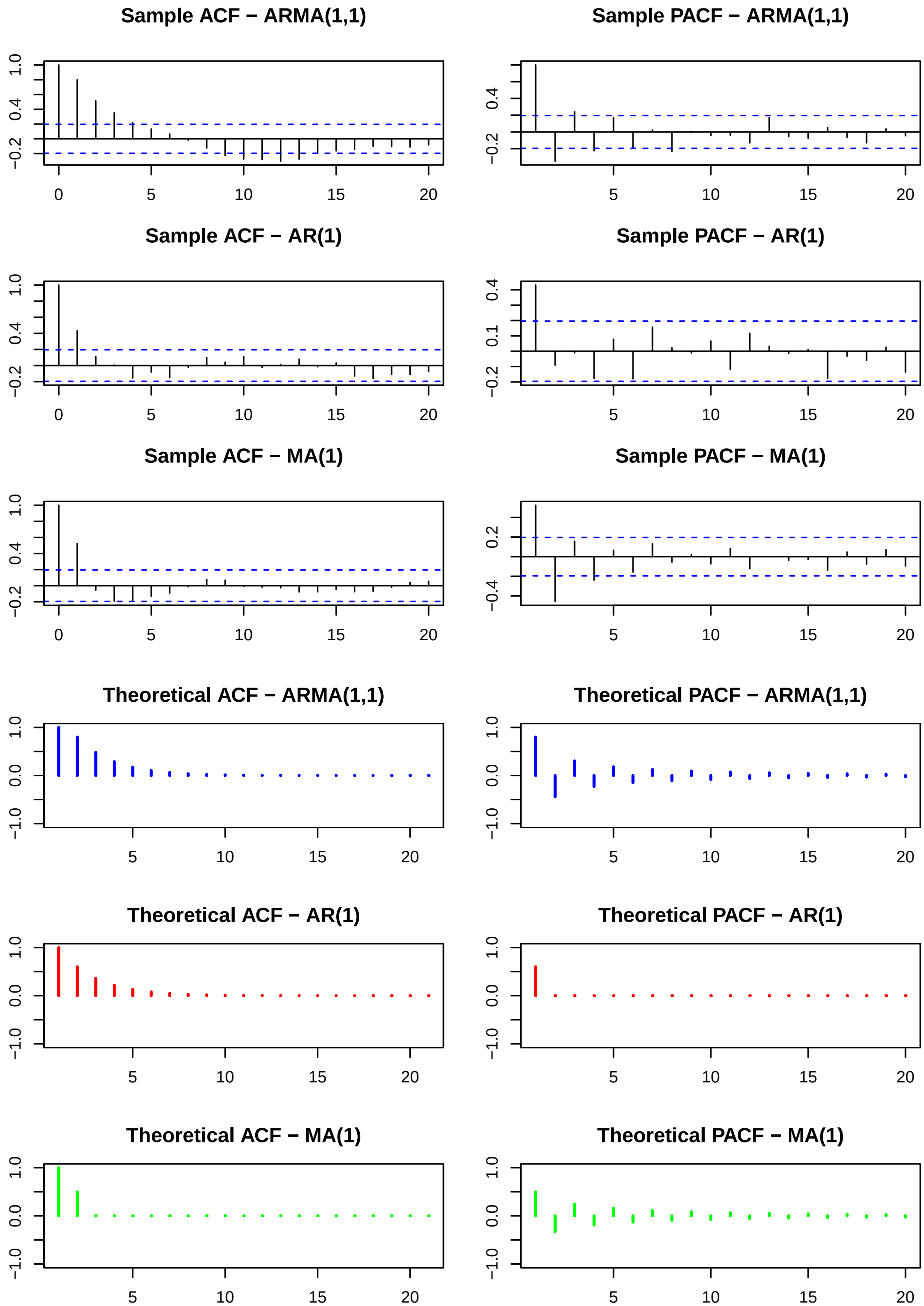
Part g

The moving average forecast drops to 0 much quicker, specifically within only two months. This occurs because a moving average model directly depends on the q prior terms with no effect from any of the terms before that, whereas an autoregressive forecast indirectly depends on all of the terms before it because each of the p lagged Xs depend on p lagged Xs before them which depend on each of the p lagged Xs before them and so on.

Part h

The new model would be $Z_t = \phi_1^3 Z_{t-1} + W_t'$ where $W_t' = N(0, \sigma^2 + \sigma^2\phi_1^2 + \sigma^2\phi_1^4)$.

Problem 2


Sample ACF − ARMA(1,1)
Sample PACF − ARMA(1,1)
Sample ACF − AR(1)
Sample PACF − AR(1)
Sample ACF − MA(1)
Sample PACF − MA(1)
Theoretical ACF − ARMA(1,1)
Theoretical PACF − ARMA(1,1)
Theoretical ACF − AR(1)
Theoretical PACF − AR(1)
Theoretical ACF − MA(1)
Theoretical PACF − MA(1)

We can see that sample ACFs roughly align with what is expected: The ACF and PACF of the ARMA(1,1) both tail off, the ACF of the AR(1) tails off but the PACF collapses after lag 1, and the ACF of the MA(1) collapses after lag 1 but the PACF tails off.

Problem 3

The general form of the m step ahead forecast is $x^t_{t+m} = \phi^m x_t$.

$$x_{t+m} = \phi^m x_t + \sum_{i=0}^{m-1} \phi^i w_{t+m-i}.$$

$$e^t_{t+m} = x_{t+m} - x^t_{t+m} = \sum_{i=0}^{m-1} \phi^i w_{t+m-i}.$$

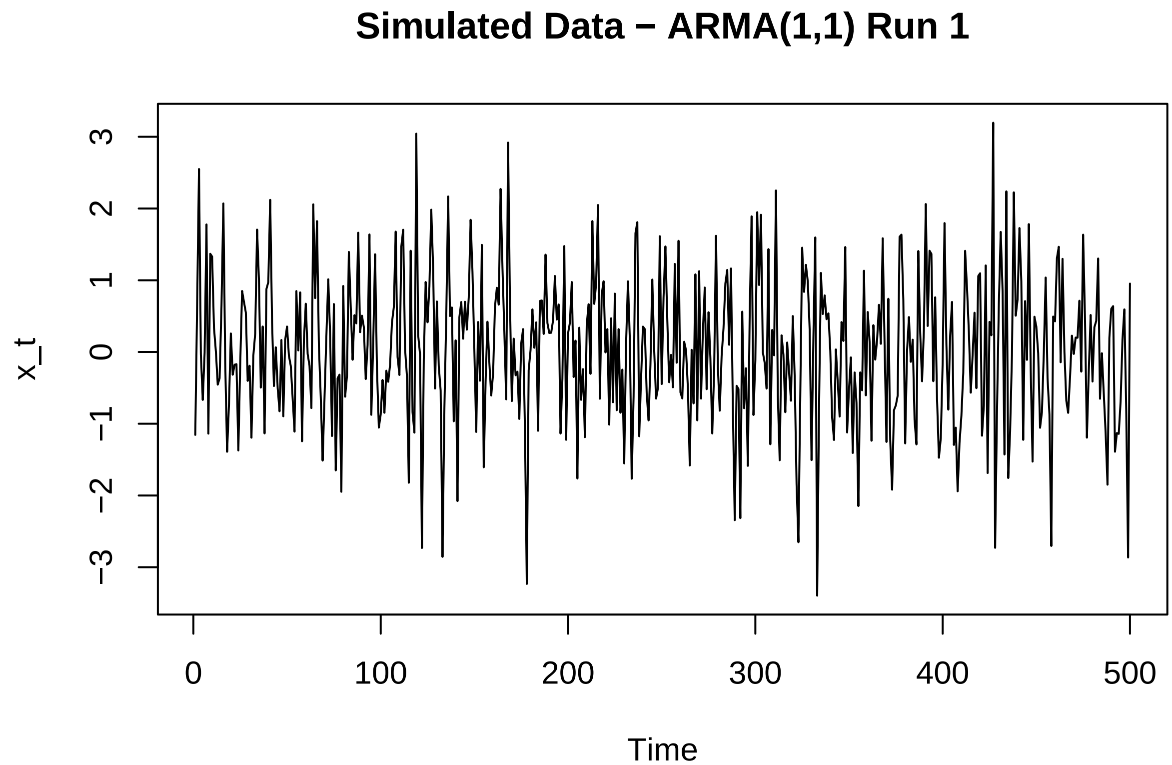
$$E[(x_{t+m} - x^t_{t+m})^2] = E\left[\left(\sum_{i=0}^{m-1} \phi^i w_{t+m-i}\right)^2\right].$$

$$E[(x_{t+m} - x^t_{t+m})^2] = \sum_{i=0}^{m-1} (\phi^i)^2 \sigma_w^2 = \sigma_w^2 \sum_{i=0}^{m-1} \phi^{2i}.$$

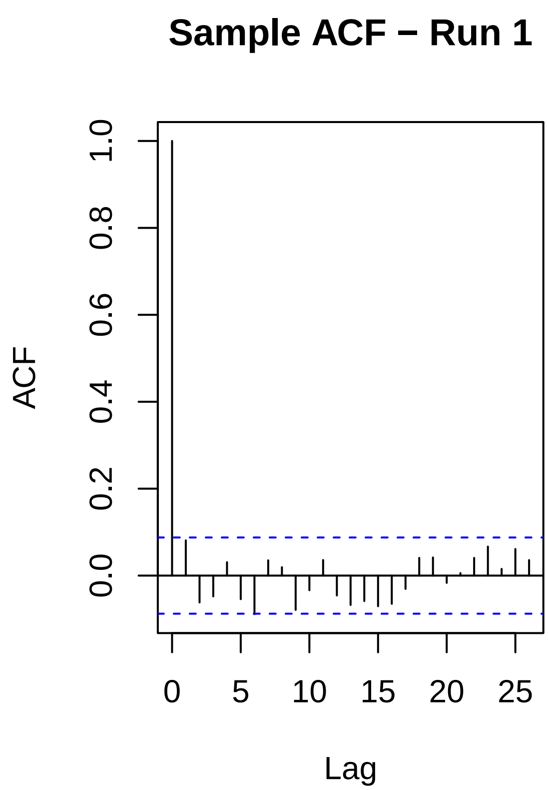
$$\sum_{i=0}^{m-1} \phi^{2i} = \frac{1-\phi^{2m}}{1-\phi^2}.$$

$$E[(x_{t+m} - x^t_{t+m})^2] = \sigma_w^2 \frac{1-\phi^{2m}}{1-\phi^2}.$$

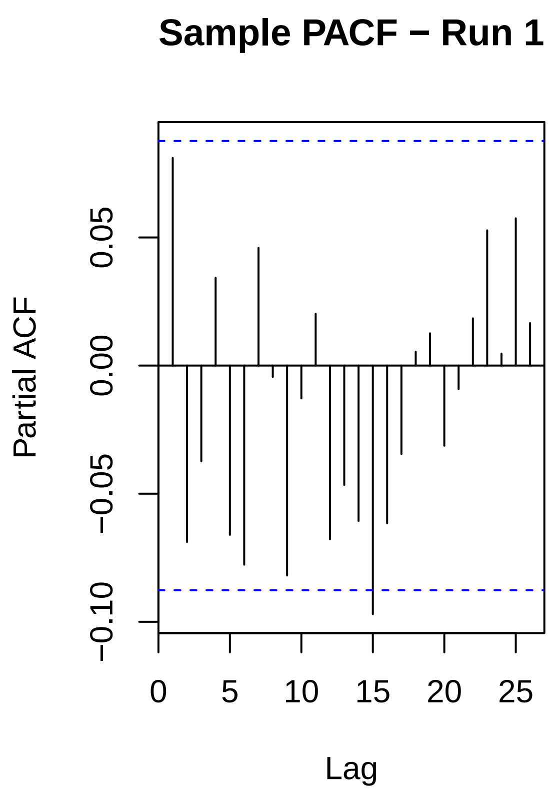
Problem 4



Simulated Data − ARMA(1,1) Run 1
x_t
Time

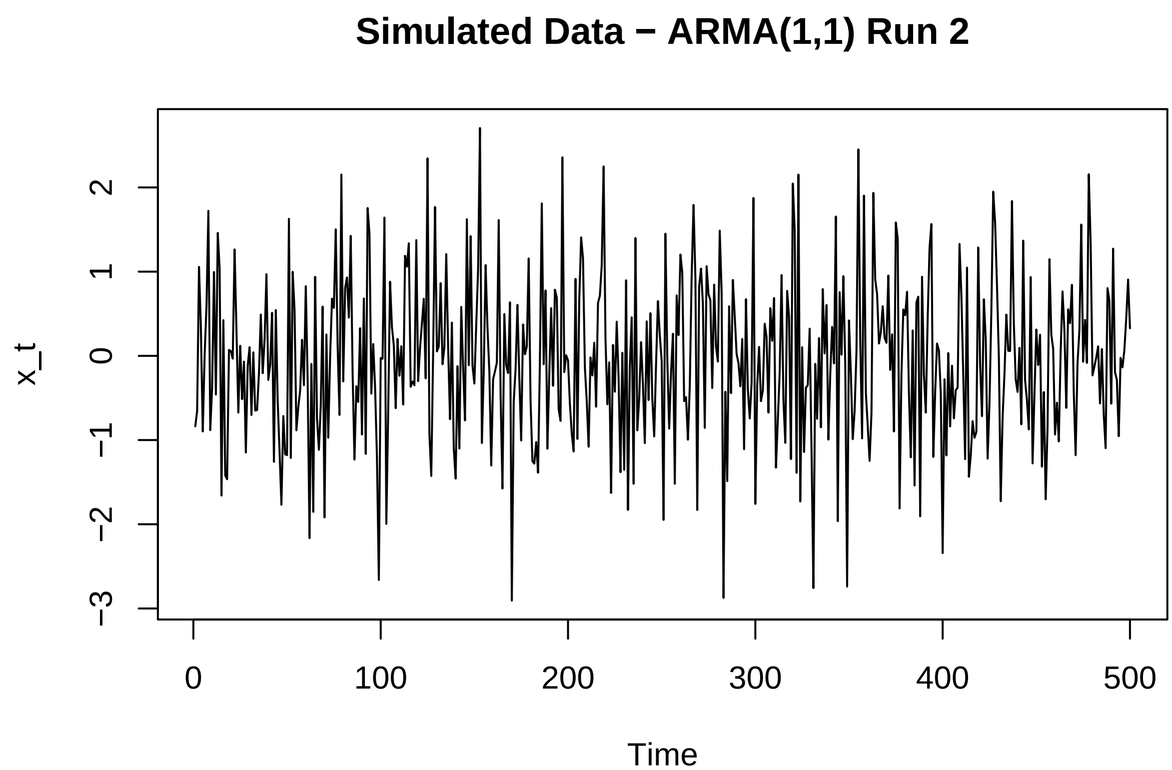


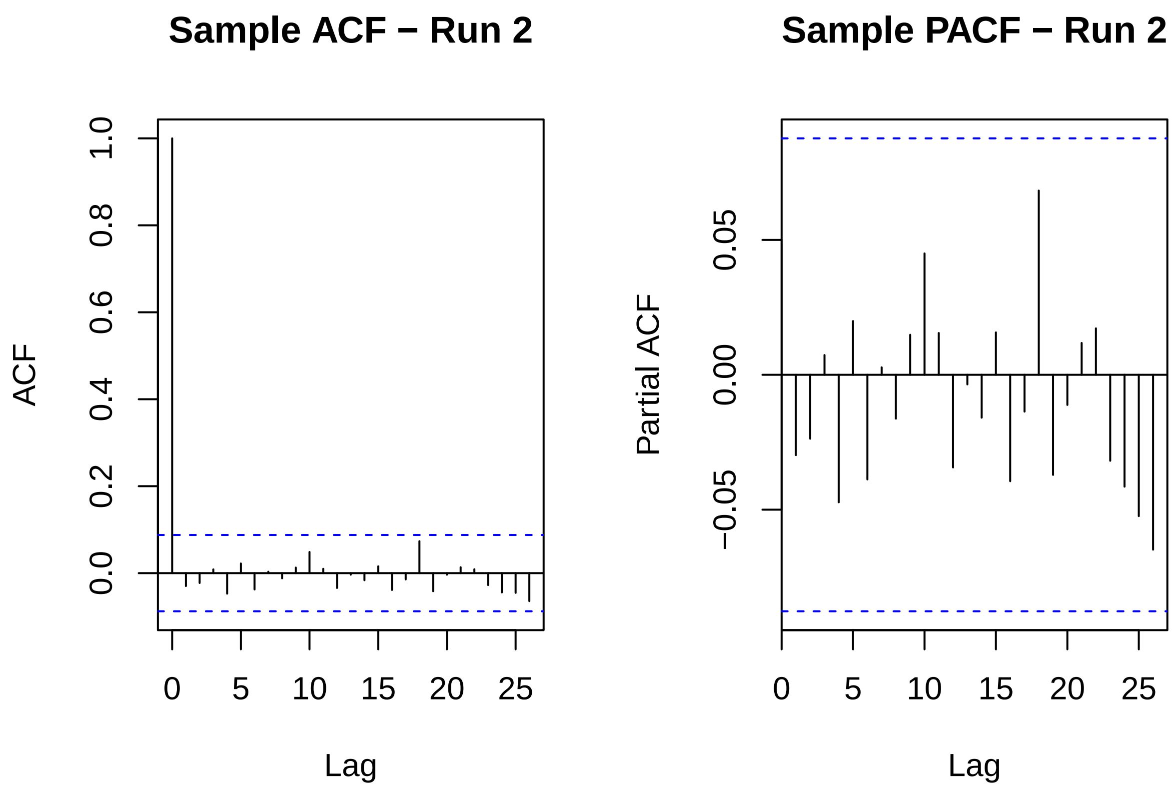
Sample ACF − Run 1
ACF
Lag



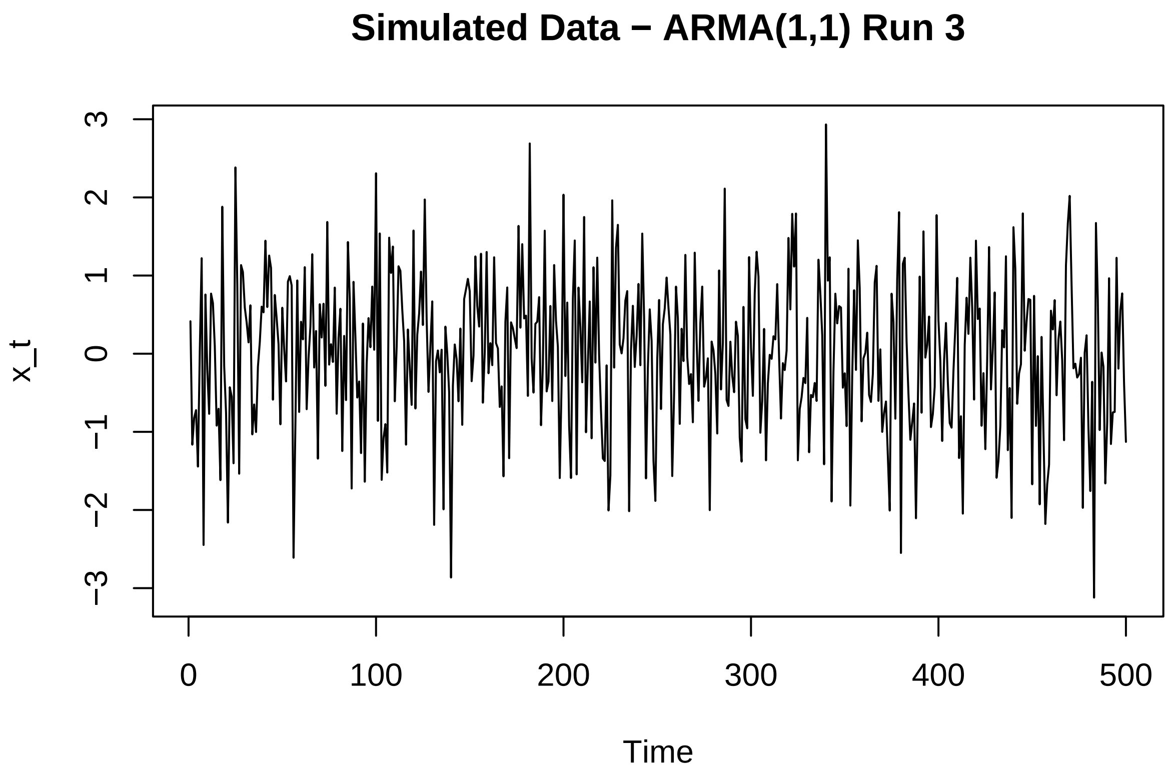
Sample PACF − Run 1
Partial ACF
Lag

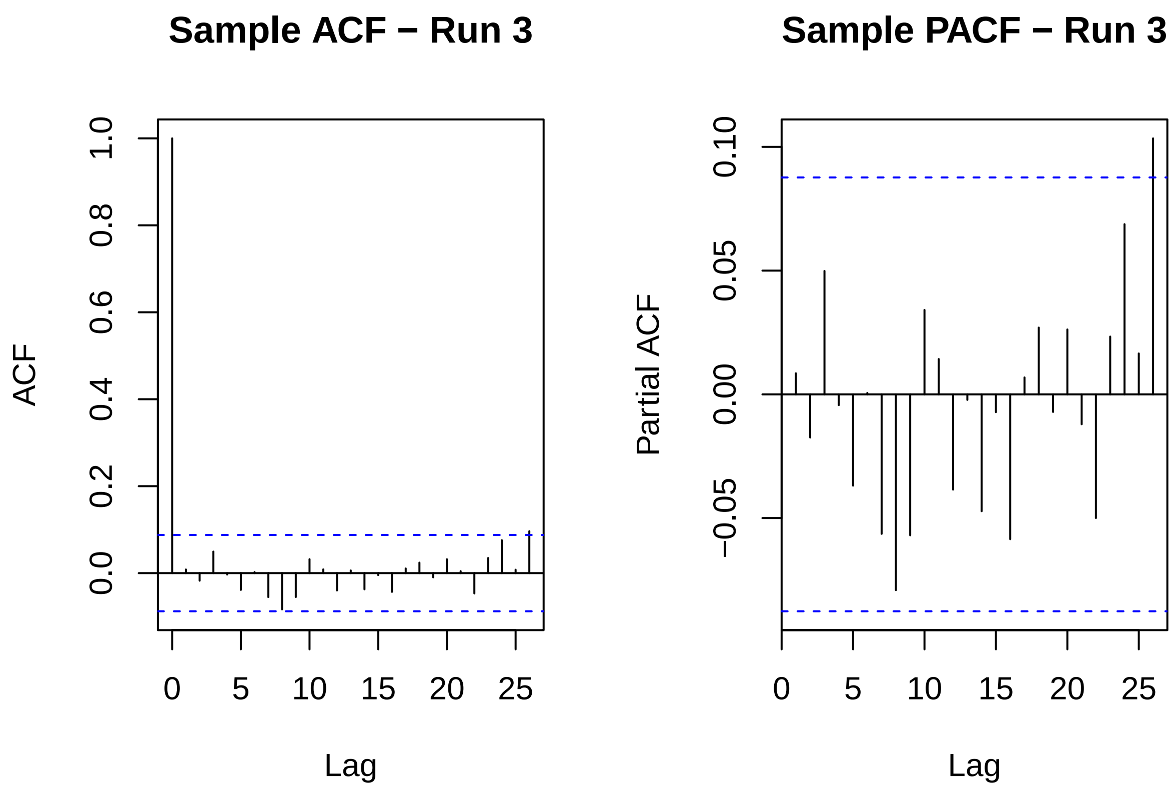
```
##
## Results for Run 1 :
##
## Call:
## arima(x = arma_data, order = c(1, 0, 1))
##
## Coefficients:
##           ar1     ma1  intercept
##       -0.2173  0.3085     0.0387
## s.e.   0.2837  0.2746     0.0492
##
## sigma^2 estimated as 1.049:  log likelihood = -721.45,  aic = 1450.91
```






```
##
## Results for Run 2 :
##
## Call:
## arima(x = arma_data, order = c(1, 0, 1))
##
## Coefficients:
##          ar1      ma1  intercept
##       0.6536  -0.6848    -0.0497
## s.e.  1.1262   1.0916     0.0385
##
## sigma^2 estimated as 0.8923:  log likelihood = -680.99,  aic = 1369.98
```


Simulated Data − ARMA(1,1) Run 3
x_t
3
2
1
0
−1
−2
−3
0
100
200
300
400
500
Time




```
##
## Results for Run 3 :
##
## Call:
## arima(x = arma_data, order = c(1, 0, 1))
##
## Coefficients:
##           ar1     ma1  intercept
##       -0.7257  0.7422    -0.0138
## s.e.   0.8023  0.7678     0.0441
##
## sigma^2 estimated as 0.9544:  log likelihood = -697.8,  aic = 1403.6
```

This ARMA model has a redundant root (.9), so it is actually an ARMA(0,0) instead of an ARMA(1,1), meaning it's just white noise. Thus, the estimates for ϕ and θ are very different for each run, as they are not based on data from and ARMA(1,1) process but rather white noise.

Problem 5

Part a

These results indicate that the time series likely isn't stationary because one would not expect such large changes in parameters from removing the last 100 observations if the mean and variance were not changing with time.

Part b

A diagnostic tool to assess the stationarity of the series is the ACF. The analyst could plot the ACF and observe whether the values decay quickly to zero. If the decay is very slow the series is likely not stationary.